



pH Rainbow Adventure!

Fun with Magic Juice, Acids, and Bases! ✨

 Made for Grade 1 Science Detectives 

Hi Teachers and Parents! This is a fun storybook science guide.

We use super simple words so your little ones can enjoy every single page!

1. Welcome, Science Detectives!

Hello, little friends! Welcome to our science lab. Today, you are not just kids—you are **Science Detectives!** 🕵️🕵️

Our world is full of colorful secrets, and today we are going to unlock one together using everyday things from your kitchen.

Have you ever seen colors change right before your eyes? It looks like real magic, but it is actually something even better. It is **Chemistry!** Chemistry is the science of mixing things to see how they work.

☀️ Your Detective Goal

Today, we will learn how to spot secret clues using our eyes, our tongues, and our fingers. We will make a beautiful rainbow inside real glass cups! Are you ready to explore? Let's go!

2. The Mystery Lunch Stain! 🤯

Let's start our first mystery story. Imagine you are sitting at the lunch table, eating a yummy, warm **Aloo Paratha** flatbread. Yum! 😋

Uh oh! A tiny piece drops onto your clean white shirt. It leaves behind a bright, sunny **yellow stain!** That yellow color comes from a special kitchen spice called **Turmeric**.

You run to the sink to clean it. You take some soapy water and rub the spot. But wait! Something crazy happens!

The yellow stain doesn't disappear. It doesn't fade away. Instead, it turns **BRIGHT RED!** 🛑

Wow! Why did a yellow spot turn red just from touching soap? That is our big mystery for today!

3. The Super Safety Rule!

Before we solve our mystery, we need to learn the most important rule in the whole wide world of science.

Science tools can be lots of fun, but some liquids can make our eyes sting or boo-boos on our skin. So, real scientists always follow the **Golden Rule!**

Let's practice our secret scientist chant together. Stand up, put your hands on your hips, and say it out loud:

**"No-Sip, No-Sniff!
Always ask an adult!"** 

This means:

- **No-Sip:** We NEVER put lab liquids in our mouths. No drinking!
- **No-Sniff:** We NEVER put our noses close to smell the cups.
- **Ask an Adult:** We only mix things when a grown-up helper says it is okay!

4. Meet the Acid Family! 🍊

Now, let's travel to a magical place called **Kingdom Acid!** 🏰

The friends who live here are called the **Acid Family**. They are super fast, energetic flavor heroes! Let's meet two of them:

- **Lucy Lemon:** She is bright yellow and loves to make funny faces!
- **Tamarind Water:** A tangy, brown plant juice friend.

🍷 **How to Spot an Acid:**

Acids have a big secret: they taste super, super **SOUR!**

Can you show your teacher your best "sour face"? Squeeze your eyes shut tight, wrinkle up your nose, and tuck in your chin! That puckering feeling in your mouth tells you that you just met an acid!

5. Acid Magic Colors!

The Acid Family has a favorite color game. They absolutely love bright, happy colors like **pink and red!**

In our lab, we have a special purple potion made from boiled red cabbage leaves. We call it our **Magic Juice!** 

What happens when an acid meets the Magic Juice? Let's find out!

If you take a dropper, scoop up some purple Magic Juice, and squeeze it into a cup of sour lemon juice... ZIP! 

The dark purple liquid transforms right before your eyes into a beautiful, glowing **bright pink-magenta!**

Whenever Magic Juice turns pink, it is a secret code that says: *"An Acid friend is hiding in this cup!"*

6. Meet the Base Family!

Next, let's float over to a peaceful place called **Kingdom Base!** 

The friends who live here are called the **Base Family**. They are very calm, slow, and smooth. Let's meet them:

- **Sammy Soap:** Our bubbly friend who loves to clean your hands!
- **Whitewash Water:** A chalky helper that paints garden walls white.

How to Spot a Base:

Bases have a completely different secret clue: they feel super, super **SLIPPERY!**

Let's try it: rub your gloved thumbs and fingers together smoothly in circles. Feel how soft and slick that is? It slides around like ice cream melting on a warm day. That slippery feeling means you found a base! Bases also taste very **bitter** (but remember our safety rule: no tasting!).

7. Base Magic Colors!

The Base Family doesn't like pink or red at all. They think those colors are way too loud!

Instead, the Base Family loves cool, calm colors like **blue and green**.  

Let's see what happens when a base touches our purple cabbage Magic Juice.

Take your dropper, gather some purple Magic Juice, and drop it into a cup of soapy water. Splash! 

The purple juice changes instantly into a deep, rich **forest green or pretty ocean blue!**

Whenever Magic Juice turns green or blue, it is a secret code that says: *"A slippery Base friend is resting in this cup!"*

8. Time for our Live Lab Show!

Now it is your turn to make a real chemical rainbow! Follow these easy steps with your adult helper:

The Detective Steps:

1. Line up **three clear cups** in a straight row on your tray.
2. Pour a splash of sour **lemon juice** into Cup 1.
3. Pour a splash of plain, clean **water** into Cup 2.
4. Pour a splash of slippery **soap water** into Cup 3.
5. Use your pipette dropper to add 3 drops of purple **Magic Juice** into every cup! Give them a gentle swirl.

Watch the Rainbow Appear!

- **Cup 1 (Lemon):** Boom! It turns **VIVID PINK!**
- **Cup 2 (Water):** Wait! It stays **PURPLE!** Water is neutral, which means it is a peaceful helper that does not change the color.
- **Cup 3 (Soap):** Wow! It turns **DEEP GREEN!**

9. The Secret Rainbow Code!

Great job, Detectives! You just made a beautiful color rainbow. Let's look at our secret scientist chart to remember what the color codes mean:

Family Name	Secret Clue	Magic Juice Color	Kitchen Friends
Acid Family	Tastes Sour 🤢	Pink / Red	Lemon, Tamarind, Sour Curd
Neutral Water	No Taste 💧	Purple	Plain Clean Water
Base Family	Feels Slippery 🧼	Green / Blue	Hand Soap, Detergent

Now you can look at any liquid and know exactly who is hiding inside just by looking at the color!

10. Scientist Detective Quiz!

Let's see if you can solve these two fun science riddles! Look at the choices below and point to the right answer.

Riddle Number 1: The Sour Curd Bowl

You find a bowl of white yogurt or sour curd in the kitchen. It tastes very tangy and sour. If we add our purple Magic Juice into it, what color will it turn?



A. Bright Pink (Because it is a sour acid!)



B. Ocean Green (Because it is a slippery base!)

Answer Key: The correct choice is **A!** Since curd is sour, it belongs to the Acid Family, and acids love pink!

Riddle Number 2: The Soap Splash

What color does Magic Juice show when it mixes with slippery soap water?

Answer: It turns **Green or Blue!** (Choice B).

11. Your Home Mission: The Yellow Code!

Now it is time to take your science powers home! Remember our big mystery from lunch? The yellow turmeric stain that turned red when touched by soap?

You can recreate that exact magic color change tonight to surprise your family!

Your Mission Checklist:

- **Step 1:** Ask a grown-up helper to put half a spoon of yellow turmeric powder into a glass of warm water. Stir it up until it looks like golden sunshine! ☀️
- **Step 2:** In another cup, mix some liquid hand soap with water until it is nice and bubbly. 🫧
- **Step 3:** Pour the soapy water into the yellow turmeric cup. Watch closely... WOW! It turns **BLOOD RED!** 🛑

⚠️ Warning for Little Scientists:

Turmeric is a very strong yellow dye. It can stain kitchen counters or clothes permanently! Make sure you do this fun experiment right over the kitchen sink with an adult watching you the whole time.

🌈 High five, Science Detectives! Keep exploring the beautiful, colorful world around you! 🎉